Imperfect Competition III: Product Differentiation

1 Definition

1. Vertical differentiation. Different products correspond to different qualities of the same goods.

2. Horizontal differentiation. Consumers show differences in subjective tastes.

3. A model of vertical differentiation. Consider a good with two qualities: $s_1 < s_2$. The price of the better quality is higher $p_1 < p_2$. The consumers' utility function is $U = \theta s - p$, where θ is a quality preference parameter. The consumers must therefore compare $\theta s_1 - p_1$, $\theta s_2 - p_2$,and 0.

4. A model of horizontal differentiation. Hotelling (1929) proposed the first horizontal differentiation model. It is still the most used model today.

5. Imagine a linear city as segment [0, 1]. There are two stores each located at an extremity of the city. Their prices are p_0 and p_1. The consumers are uniformly spread throughout the city. Consumers face a transport cost t per unit of distance covered.

6. There are two possible models: (1) of travel costs incurred by consumers in order to reach each store; (2) of transport costs incurred by the stores in delivering products to consumers.

7. The product present itself in numerous varieties indexed by a parameter of [0, 1]. Company 0 chooses to sell variety 0 and firm 1 variety 1. The consumer situated in x prefers variety x and must discharge a cost (in utility) $t\left|x - x'\right|$ if he buys variety x'.

8. The consumer located in x perceives a generalized price, $p_0 + tx$, for the store situated in 0, and thus $p_1 + t(1-x)$ for the store situated in 1. If P is the subjective value of the good for each consumer, the demand aimed at store 0 is given by the smallest x such that

$$\begin{aligned} p_0 + tx &\leq p_1 + t(1-x) \\ p_0 + tx &\leq P. \end{aligned}$$

9. It is then

$$\begin{array}{c} \dfrac{p_1 + t - p_0}{2t} \text{ if } \dfrac{p_1 + t + p_0}{2} \leq P \\ \dfrac{P - p_0}{t} \text{ if } \dfrac{p_1 + t + p_0}{2} < P \text{ and } p_0 \leq P \\ 0 \qquad\qquad \text{if } p_0 > P. \end{array}$$

2 Differentiation and Monopoly

1. Take now a vertically differentiated good of quality index s. Suppose that the inverse demand function of the consumers is $P(q, s)$, which increase in s. The production cost $C(q, s)$ also rises with the quality.

2. The monopolist chooses the quantity and the quality so that

$$\max_{q,s} \, [qP(q, s) - C(q, s)].$$

3. Now look more realistically at a situation where only the seller knows the quality s of his product. The buyer only knows that s is uniformly distributed on [0,1]. The seller's utility is $p - \theta_1 s$, and that of the single buyer is $\theta_2 s - p$, where θ_1 and θ_2 are two publicly known parameters such that $\theta_1 < \theta_2$.

4. Hence, it is socially optimal for the buyer to acquire any quality from the seller. Still we will see that the unobservability of quality can cause a market breakdown.

5. When a seller proposes a price p, the buyers could reason as follows: if the seller can sell at p, it is that $p \geq \theta_1 s$. There are two possibilities:

   (a) $p \geq \theta_1$. In this case I learn nothing of the quality by observing the price. So I must evaluate the average quality of good for sale at $\frac{1}{2}$, and the average utility that I can have in purchasing is therefore $\theta_2/2 - p$. If $\theta_2 < 2\theta_1$, this utility is then negative, so I will not buy.

(b) $p < \theta_1$. The average quality that I can expect is $p/2\theta_1$, which gives me an average utility $p\theta_2/2\theta_1 - p$. This utility is still negative if $\theta_2 < 2\theta_1$.

6. The conclusion is immediate: if $\theta_1 < \theta_2 < 2\theta_1$, no quality of the good can be sold, even though a sale should be socially optimal. This situation is often called the *lemon problem* after a famous example of Akerlof (1970).

7. The *lemon problem* is fortunately not irremediable; it can be solved by introducing a signal of quality for a good. For example, introducing guarantees can establish a more satisfying (albeit second-best) equilibrium: only the sellers of good cars will offer a guarantee, since it would be costly for sellers of bad cars to do so. This is the famous costly signaling device.

8. Now, in a horizontal differentiation model, would a monopoly introduce too many or too few products with relation to the social optimum? There are two opposing arguments:

   (a) A monoply can restrain the variety of its products. In the absence of first-degree price discrimination, a monopoly's profit π is less than the social surplus S, which also comprises consumer surplus. A monopoly will not introduce a new product if its fixed cost f is between π and S, though the product is socially desirable.

   (b) Monopoly prices above the marginal cost on a product can make other varieties of the product more interesting for consumers. The monopoly can therefore gain by introducing new varieties even if not socially optimal.

9. We examine the second argument with Hotelling model. In this model, the social surplus maximization reverts to minimizing the sum of the transport costs and firms' production costs (the prices paid are merely transfers between consumers and firms).

10. Suppose that the marginal costs are 0 but that opening a store costs f. If there is only one store, situated at 0, the total transport costs are given by $\int_0^1 tx dx = t/2$; if a new store is open at 1, the transport costs become $2\int_0^{\frac{1}{2}} tx dx = t/4$. If the cost of creating a

new store f is superior to $t/4$, it is therefore socially preferable to open only one store.

11. How about a monoply? Suppose that the subjective value of the good P is high so that the whole market is covered. Then the monopoly will tariff so as to leave a 0 surplus to the most distant consumer: $p = P - t$ if there is only one store, and $p = P - t/2$ if there are two. The opening of a new store reduces the transport costs of consumers and thereby allows the monopoly to raise its prices.

12. Since the monopoly's profit (outside of opening cost) is simply $\pi = p$, it will prefer opening two stores provide that $f < t/2$. When $t/4 < f < t/2$, the monopoly will indeed open one more store than a social planner.

3 Differentiation and Oligopoly

1. When there are several firms, one must take into account their differentiation strategies. Consider the Hotelling model with two firms, and three new hypothesis:

   (a) P is very high, and consumers then always buy

   (b) the marginal production costs are constant and equal to c

   (c) the transport costs are quadratic.

2. If the prices are fixed in the quadratic costs model, the firms will gain by drawing nearer the center of the segment: it is the *minimal differentiation principle* stated by Hotelling (1929): buyers are confronted everywhere with an excessive sameness.

3. To see this, assume that the prices practiced by the two firms are equal to p, that one firm is located in a and the othe in $1 - b$, with $a \leq 1/2 \leq 1 - b$. So the limit of markets is given by $(x - a)^2 = (1 - b - x)^2$, being $x = (a + 1 - b)/2$. The first firm's profit is px, and the profit increases in a. So its dominant strategy is to be established in $1/2$.

4. The same reasoning also brings the second firm to set up in $1/2$. We have indeed a matter of "excessive sameness," since the social optimum is to minimize the total transport costs

$$\int_0^{(a+1-b)/2} (x-a)^2 dx + \int_{(a+1-b)/2}^{1} (1-b-x)^2 dx = \frac{(1 - b - a)^3 + a^3 + b^3}{3}.$$

5. From the expression above we easily see that the costs are minimum for $a = b = 1/4$.

6. We find, however, that this principle must be inverted when the prices are freely chosen by the firms (d'Aspremont, Gabszewicz, and Thisse, 1979). In a first stage the two firms are free to localize themselves wherever they wish on the segment [0,1]. So we can consider this as two-stage game:

   (a) The firms choose their location a and $1 - b$

   (b) They subsequently choose their price p_0 and p_1.

7. We want to characterize the perfect equilibrium of this game and to demonstrate that $a = b = 0$ at equilibrium, that is, that the firms seek to differentiate themselves to the maximum.

8. We start at the game's end, after the firms have settled in their location. If they tariff p_0 and p_1, the boundary x that delimits their territories is given by

$$p_0 + t(x - a)^2 = p_1 + t(1 - b - x)^2.$$

9. The demand applying to 0 is therefore

$$D_0(p_0, p_1, a, b) = a + \frac{1 - b - a}{2} + \frac{p_1 - p_0}{2t(1 - b - a)},$$

   which comprises three terms. The first is the territory situated to the left of firm 0, the second corresponds to the equal distribution of the territory situated between the two firms, and the last is the effect due to the difference in price.

10. Naturally, the demand aimed at 1 is simply $D_1 = 1 - D_0$. If each firm has a constant marginal cost c, the Nash price equilibrium is given by the maximization of $(p_0 - c)D_0$ in p_0 and $(p_1 - c)D_1$ in p_1. Simple calculation gives

$$p_0(a, b) = c + t(1 - b - a)(1 + \frac{a - b}{3})$$

    and a similar formula for p_1.

11. Now we can go to the first stage. Firm 0 will choose a to maximize

$$\pi_0 = (p_0(a, b) - c)D_0(p_0(a, b), p_1(a, b), a, b)$$

In differentiating, it gets

$$\frac{\partial \pi_0}{\partial a} = \frac{\partial \pi_0}{\partial p_0}\frac{\partial p_0}{\partial a} + (p_0 - c)\frac{\partial D_0}{\partial p_1}\frac{\partial p_1}{\partial a} + (p_0 - c)\frac{\partial D_0}{\partial a}.$$

12. The first term is 0 by the envelop theorem, since p_0 makes π_0 maximal. The second term is the strategic effect: as $\frac{\partial D_0}{\partial p_1} > 0$ and $\frac{\partial p_1}{\partial a} < 0$, to reduce a permits firm 0 to incite 1 to raise its price, and therefore firm 0 acquires more clients. The last term, as$\frac{\partial D_0}{\partial a} > 0$, simply express that being near downtown permits firm 0 to take clients from firm 1.

13. After some calculation of little interest, we can show that the strategic effect carries it away and that one therefore gets $\frac{\partial \pi_0}{\partial a} < 0$, so at equilibrium, firm 0 positioned at $a = 0$. Identical reasoning bearing on firm 1 would show that one gets $b = 0$.

14. So the two firms choose to position themselves at the two extremities of the segment. This is the maximal differentiation principle. It calls for several remarks:

    (a) The maximal differentiation principle induces too much differentiation. We saw earlier that $a = b = 1/4$ are the social optimum.

    (b) This principle is not general. For example, it is false if the transport costs are $x^{\frac{3}{2}}$.

15. Nonetheless, Hotelling model has been assembled to explain both the minimal differentiations and the maximal differentiations.

4 Entry and Numbers of Products

1. We now introduce entry and consider another model by Salop (1979). This model's theme is that there are too many products at equilibrium in relation to the social optimum.

2. Consider a circular city upon with consumers and n firms are uniformly distributed. The transport costs are supposed linear. Here again the value of the good for each consumer is assumed to be very high, so the whole market will be covered at equilibrium. Each firm has a fixed cost f and constant marginal costs c.

3. At symmetrical equilibrium, all firms tariff the same price p. A firm that chooses a different price p' can ensure itself a market of size x to both sides of itself, where x is given by

$$p' + tx = p + t(\frac{1}{n} - x).$$

4. The firm would then get a demand $(1/n + (p - p')/t)$ and a profit

$$\pi = (p' - c)(\frac{1}{n} + \frac{p - p'}{t}) - f$$

which is maximal in $p' = (p + t/n + c)/2$. The condition $p' = p$ gives the equilibrium

$$p = c + \frac{t}{n}, \ \pi = \frac{t}{n^2} - f.$$

5. At free-entry equilibrium, this profit is 0 and there are therefore $\sqrt{t/f}$ firms. The social optimum minimizes the sum of fixed costs nf and of transport costs $2n \int_0^{1/2n} tx dx = t/4n$, which this gives $n = \frac{1}{2}\sqrt{t/f}$. Clearly, there are too many firms at free-entry equilibrium, because of business stealing: a firm that decides to enter the market takes into account only its own profit and not the decrease of profits of firms already there.

6. Among the factors that can explain how firms that compete against each other by price can sustain supranormal profits, barriers to entry hold an important place. Several different types of barriers to entry can exist:

   (a) Legal barriers. In certain countries the law restrict the number of operators in certain professions. Other industries are ruled by permit system.

   (b) Scale economies.

   (c) Strategic barriers.

   (d) Technologic barriers. A firm with technological advantages at its disposal, e.g. an accumulation of R&D, can make entry difficult for competitors.

5 Homework

1. Consider a Cournot duopoly in which the firms have a constant marginal cost c and the inverse demand function is $p(q) = a - bq$, with $a > c \geq 0$ and $b > 0$.

(a) What are the socially optimal (competitive) output and price? What are the jointly monopoly quantity and price, what is the profit for each firm suppose they divide the monopolistic profit equally?

(b) What are the Cournot competition outcome (quantity, price and profit for each firm)?

(c) Suppose each firm can choose "collusion" or "deviation". If one firm colludes, the other firm deviates, what is the outcome? What is the Nash equilibrium for the static (or one-shot) game?

(d) Now consider the infinitely repeated Cournot duopoly with discount factor $\delta < 1$. Under what conditions can the monopoly (or collusive) outcome be supported as an equilibrium outcome? Specify the equilibrium strategy.

2. Consider the widget market. The total demand by men for widgets is given by $x_m = a - \theta_m p$, and the total demand by women is given by $x_w = a - \theta_w p$, where $\theta_w < \theta_m$. The cost of production is c per widget.

   (a) Suppose the widget market is competitive. Find the equilibrium price and quantity sold.

   (b) Suppose, instead, that firm A is a monopolist of widgets. [Also make this assumption for (c) and (d)]. If firm A is prohibited from "discriminating" (i.e., charging different prices to men and women), what is its profit-maximizing price? Under what conditions do both men and women consume a positive level of widget in this solution?

   (c) If firm A has produced some total level of output X, what is the welfare-maximizing way to distribute it between men and women?

   (d) Suppose that firm A is allowed to discriminate. Calculate the first-, the second-, and the third-degree price discrimination results, respectively.

3. Consider a two-firm Cournot model with constant returns to scale but in which firms' costs may differ. Let c_j denote firm j's cost per unit of output produce, and assume that $c_1 > c_2$. Assume also the inverse demand function is $p(q) = a - bq$, with $a > c_1$.

(a) Derive the Nash equilibrium of this model. Under what conditions does it involves only one firm producing? Which will this be?

(b) When the equilibrium involves both firms producing, how do equilibrium outputs and profits vary when firm 1's cost changes?

(c) Now consider the general case of J firms. Show that the ratio of industry profits divided by industry revenue in any (pure strategy) Nash equilibrium is exactly H/ε, where ε is the elasticity of the market demand curve at the equilibrium price and H, the Herfindahl index of concentration, is equal to the sum of the firms' squared market shares $\sum_j (q_j^*/Q^*)^2$.

(d) Show that under repeated interactions, the collusion results can be sustained with the two-firm case.